

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 38)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

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01 . 2

Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

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01 . 3

Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

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10

- 0 2** Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

- 0 2** . **1** Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

- 0 2** . **2** Explain the purpose of aeration (stage 3).

[2 marks]

- 0 2** . **3** Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

0 2 . 4 Describe how each of the following contaminants is removed from water. **[3 marks]**

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

10

0 3 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 3 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers. **[5 marks]**

0 3 . 2

Explain why readings should be taken with the same wind direction.

[1 mark]

0 3 . 3

Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

0 3 . 4

Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

1

2

10

0 4

A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 4 . 1

Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

04 . 2

Give **three** limitations of kick sampling.

[3 marks]

1

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2

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3

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04 . 3

Explain how a Surber sampler improves on kick sampling.

[2 marks]

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04 . 4

Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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05

A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

05 . 1

Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 5 . 2

Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

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0 5 . 3

Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 5 . 4

Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

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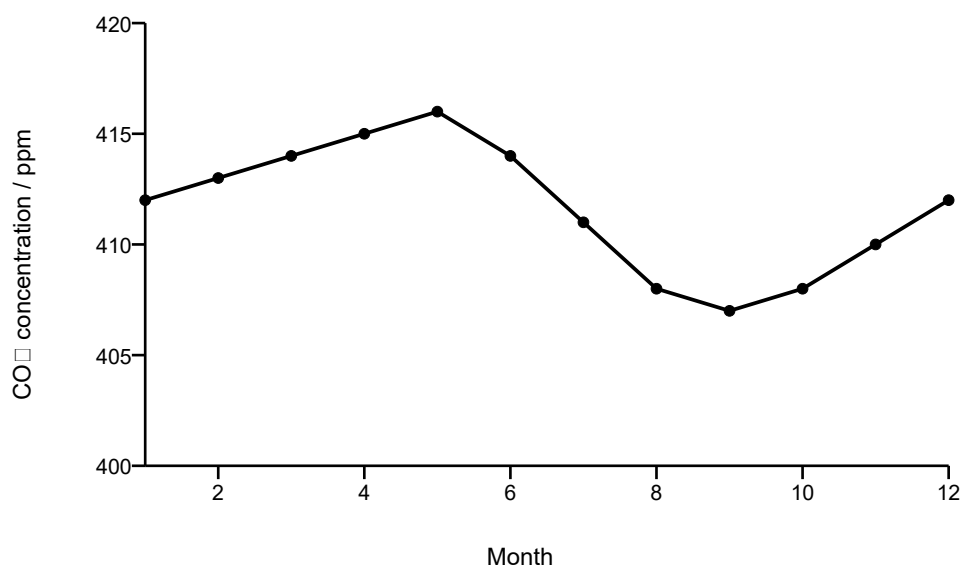
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0 6

Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1**0 6****. 1**

Describe and explain the pattern shown in Figure 1.

[4 marks]

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06 . 2

Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

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06 . 3

Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

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2

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06 . 4

Name the two layers of the atmosphere most affected by human activities.

[2 marks]

1

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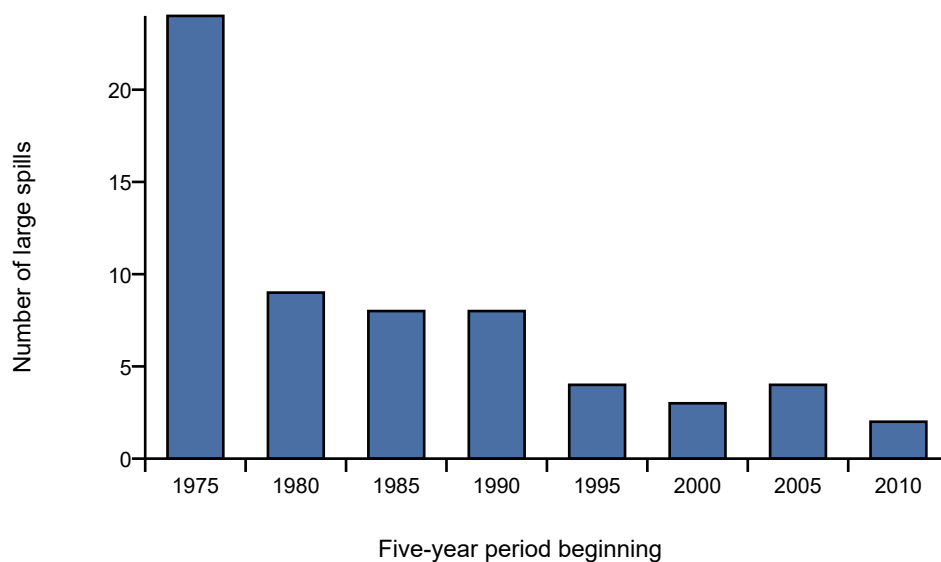
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10

07

Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1**Table 1**

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

07**1**

Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]**Answer**

%**07****2**

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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07 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1

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07 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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10

08 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

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08 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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5

09 . 1

Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

1

2

3

09 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

09 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages

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Disadvantages

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1 0 . 1

Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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Extra space

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10 . 2 State what is meant by a **carbon footprint**.

[1 mark]

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Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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A large rectangular box containing 30 horizontal dotted lines for writing.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

END OF QUESTIONS